操作一

$$\frac{d}{dx}u(x)v(x)=\frac{d}{dx}u(x)\times v(x)+\frac{d}{dx}v(x)\times u(x)$$
$$d\,u(x)v(x)=d\,u(x)\times v(x)+d\,v(x)\times u(x)$$
$$\int d\,u(x)v(x)=\int d\,u(x)\times v(x)+\int d\,v(x)\times u(x)$$
$$u(x)v(x)=\int v(x)du(x)+\int u(x)dv(x)$$

后面两个有一个能求另外一个就能求。

例：

$$\int \ln x dx=x\ln x-\int x d\ln x=x\ln x-\int x\frac{1}{x}dx=x\ln x-x+C$$
$$\int xe^x dx=\int xde^x=xe^x-\int e^x dx=xe^x-e^x+C$$
$$\int e^x\sin x dx$$
$$A=\int e^x\sin x dx=-\int e^x d\cos x=-(e^x\cos x-\int\cos x de^x)=-e^x\cos x+\int\cos xe^x dx$$
$$B=\int e^x\sin x dx=\int\sin x de^x=\sin xe^x-\int e^x d\sin x=\sin xe^x-\int e^x\cos x dx$$
$$A+B=2\int e^x\sin x dx=e^x\sin x-e^x\cos x+C$$
$$\int e^x\sin x dx=\frac{e^x}{2}(\sin x-\cos x)+C$$

总结：要么直接根据 d 前后分为 $u(x)$ 和 $v(x)$，要么先替换（如：$e^x dx\to de^x$）后利用公式。

操作二 分式函数的处理

可因式分解则因式分解，如何考虑如下形式的变换：

$$\frac{?}{(A_1x+B_1)^{k_1}(A_2x^2+B_2x+C)^{k_2}}=$$
$$\frac{q_1}{A_1x+B_1}+\frac{q_2}{(A_1x+B_1)^2}+\cdots+\frac{q_{k_1}}{(A_1x+B_1)^{k_1}}+$$
$$\frac{w_1+e_1x}{(A_2x^2+B_2x+C)}+\frac{w_2+e_2x}{(A_2x^2+B_2x+C)^2}+\cdots+\frac{w_{k_2}+e_{k_2}x}{(A_2x^2+B_2x+C)^{k_2}}$$

各个分式分开后容易积分

操作三 三角函数的处理

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = -\int \frac{1}{\cos x} d\cos x = -\ln|\cos x| + C$$
$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx = \int \frac{1}{\sin x} d\sin x = \ln|\sin x| + C$$
$$\int \sec x dx = \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx = \int \frac{d\sin x}{1-\sin^2 x} =$$
$$\frac{1}{2}\int \left(\frac{1}{1-\sin x} + \frac{1}{1+\sin x}\right) d\sin x = \frac{1}{2}(-\ln|1-\sin x| + \ln|1+\sin x|) + C$$
$$= \frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right| + C == \frac{1}{2}\ln\left|\frac{(\sin x+1)^2}{\cos^2 x}\right| + C = \ln|\sec x + \tan x| + C$$

$$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{\sin x}{\sin^2 x} dx = -\int \frac{d\cos x}{1-\cos^2 x} =$$
$$-\frac{1}{2}\int \left(\frac{1}{1-\cos x} + \frac{1}{1+\cos x}\right) d\cos x = -\frac{1}{2}(-\ln|1-\cos x| + \ln|1+\cos x|) + C$$
$$= \frac{1}{2}\ln\left|\frac{1-\cos x}{1+\cos x}\right| + C == \frac{1}{2}\ln\left|\frac{(1-\cos x)^2}{\sin^2 x}\right| + C = \ln|\csc x - \cot x| + C$$

$$\sqrt{A - Bx^2}$$
$$\sqrt{A}\sqrt{1 - \frac{B}{A}x^2} = \sqrt{1 - \left(\sqrt{\frac{B}{A}}x\right)^2}$$

可转化为 $\left(\sqrt{\frac{B}{A}}x\right) = \cos\theta$

$$\tan^2 x + 1 = \sec^2 x$$
$$d\tan x = \sec^2 x dx$$

$$\sqrt{A + Bx^2}$$
$$\sqrt{A}\sqrt{1 + \frac{B}{A}x^2} = \sqrt{1 + \left(\sqrt{\frac{B}{A}}x\right)^2}$$

可转化为 $\left(\sqrt{\frac{B}{A}}x\right) = \tan\theta$

操作四 倒代换

分母次数高 / 做不了的时候试试。

泰勒展开

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + o(x^2)$$
$$(1+f(x))^\alpha = 1 + \alpha f(x) + \frac{\alpha(\alpha-1)}{2}(f(x))^2 + o((f(x))^2)$$